210×182 TOPCon double glass single crystal half-cell module

Module type: SSA-66HD-xxxN **Power range: 600-625W**

Max power: 625W **Max efficiency: 23.14%**

Product introduction

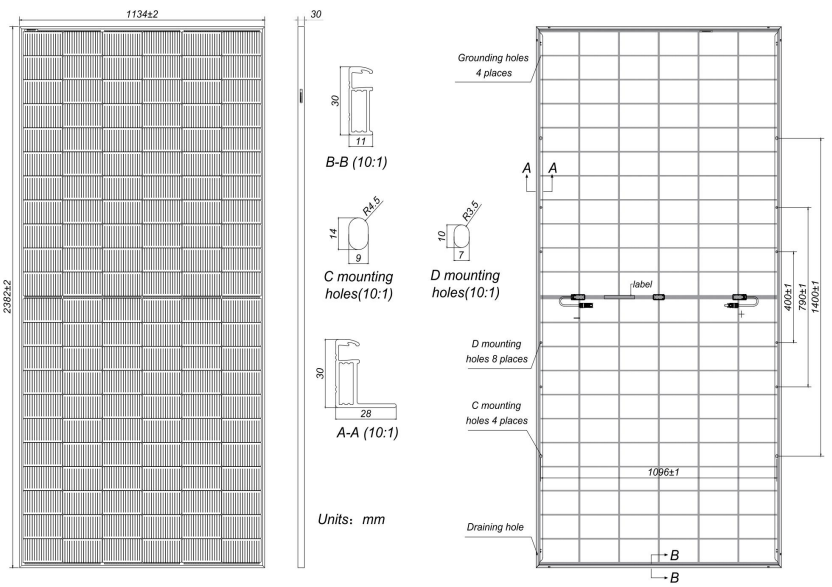
High efficiency: High efficiency TOPCon cells, superimposed multiple main gate, high-density module package, to ensure higher power output of module;

High reliability: Through third-party salt spray ammonia dust certification, the application of half cell technology ensures better resistance to heat spots and hidden cracks, improving operational reliability;

High power generation: N-type cells ensure a first-year attenuation of less than 1%, optimized circuit design brings lower shadow blocking losses and better weak light response, increasing module power generation;

High return: Optimized version design and double-sided power generation gain, effectively reduce BOS cost and LCOE cost, and improve project return (ROI);

Engineering drawing



Mechanical parameter

Cell type	TOPCon（G12R）
Cell arrangement	132(6x22)
Module size	2382×1134×30mm
Module weight	32.5kg±5%
Glass	Front, 2.0mm semi-tempered coated glass Back, 2.0mm semi-tempered white glazed glass
Frame	Anodized aluminum profile
Cable	12AWG(UL) Cable length 300mm (including connectors) or customized
Junction box	IP68, three diodes
Connector	Compatible with MC4/MC4-EVO2/ MC4-EVO2A

Electrical parameters

(STC: irradiance 1000W/m², cell temperature 25°C, spectral AM1.5G; NOCT: irradiance 800W/m², ambient temperature 20°C, spectral AM1.5G, wind speed 1m/s)

Type	SSA-66HD -600N		SSA-66HD -605N		SSA-66HD -610N		SSA-66HD -615N		SSA-66HD -620N		SSA-66HD-625N	
	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT
Max power (Pmax) [W]	600	458	605	462	610	466	615	470	620	474	625	478
Open circuit voltage (Voc)[V]	47.92	45.52	48.12	45.72	48.32	45.92	48.52	46.12	48.72	46.32	48.92	46.52
Operating voltage (Vmp) at max power point [V]	39.61	37.42	39.76	37.60	39.9	37.77	40.04	37.96	40.21	38.14	40.36	38.31
Short circuit current (lsc)[A]	15.94	12.85	16.01	12.90	16.07	12.96	16.14	13.01	16.2	13.06	16.28	13.12
Operating current at max power point (Imp) [A]	15.15	12.24	15.22	12.29	15.29	12.34	15.36	12.39	15.42	12.43	15.5	12.48
Module efficiency [%]	22.21		22.40		22.58		22.77		22.95		23.14	
Power tolerance	0~+5W											
Temperature coefficient of short circuit current (a_lsc)	+0.045%/°C											
Open circuit voltage temperature coefficient (3-Voc)	-0.260%/°C											
Maximum power temperature coefficient (Y_Pmp)	-0.290%/°C											
Rated cell operating temperature	45±2°C											
Operating temperature	-40°C~+85°C											

Back power gain

(Take 610W as an example)

Power gain	5%	10%	15%	20%	25%
Max power (Pmax) [W]	641	671	702	732	763
Open circuit voltage (Voc)[V]	48.32	48.32	48.32	48.34	48.34
Operating voltage (Vmp) at max power point [V]	39.9	39.9	39.9	39.92	39.92
Short circuit current (lsc)[A]	16.89	17.68	18.49	19.29	20.09
Operating current at max power point (Imp) [A]	16.07	16.82	17.60	18.34	19.12

Application parameter

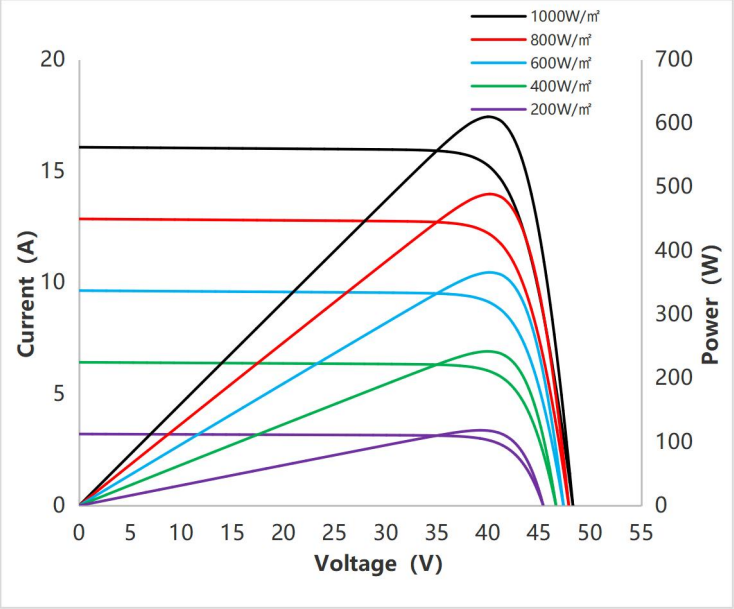
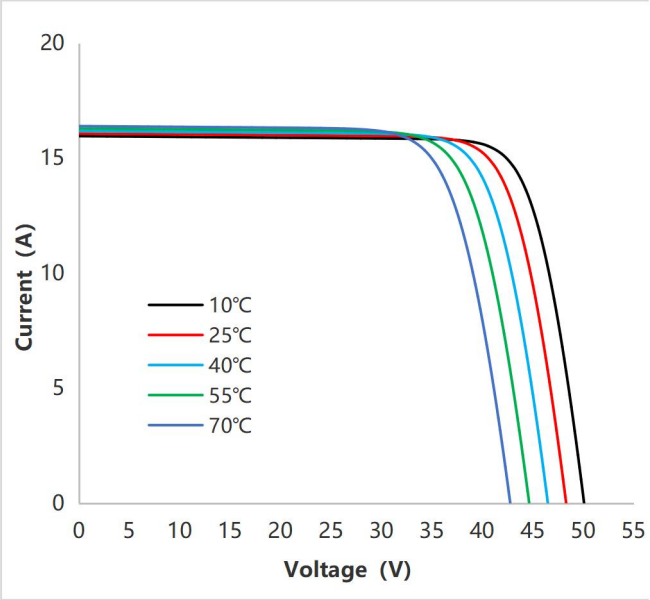
Max system voltage	1500V DC（IEC）
Max rated fuse current	30A
Max static load, front	5400Pa
Max static load, back	2400Pa
Safety level	Class II
Fire resistance	UL Type 29
Bifaciality	80%±10%

Packaging information

Vehicle type	40HQ	Flatcar（13m）
Total slice	720	720
Tablet/box	36	36
Carton number	20	20
Packing box size (mm)	2390*1130*1270	
Net weight per tow (kg)	1170	
Single drag weight (kg)	1240	

Characteristic curve chart (610W)

Current/Power Voltage Curve SSA-66HD-610N



Current-Voltage Curve SSA-66HD-610N